

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I N. Tharun Sai / 192472382 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks(100)
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	

		<i>relevant examples</i>				
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

2. A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

4. A low-contrast image requires uniform intensity distribution.

Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

5. An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

6. A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

7. An image requires noise reduction without significant loss of details.

Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

9. Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

10. A noisy image must be pre-processed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

1. An image appears too bright. Explain and justify the most suitable grey level transformation.

Introduction

Image enhancement is one of the most important operations in digital image processing. Its purpose is to improve the visual appearance of an image so that important information can be observed more clearly. A bright image usually contains very high intensity values, making it difficult to distinguish objects and fine details. To overcome this problem, a suitable grey level transformation such as gamma correction is applied to reduce the excessive brightness while preserving the essential features of the image.

Explanation

The most suitable transformation is the **Negative Transformation** or **Log Compression (depending on the brightness level)**. In most cases, **Power-Law (Gamma Correction with $\gamma > 1$)** is preferred.

Formula:

$$s = c \times r^\gamma$$

Where:

- r = input pixel value
- s = output pixel value
- $\gamma > 1$

Application

- Reduces excessive brightness.
- Makes objects more visible.
- Produces balanced illumination.

Justification

Gamma values greater than 1 compress high-intensity pixels, making the image darker and improving visibility.

Conclusion

Gamma correction is an effective grey level transformation for correcting overly bright images. By reducing the intensity of bright pixels, it improves the overall balance of the image and makes hidden details easier to observe.

2. A dark image lacks detail. Describe and justify an appropriate grey level transformation.

Introduction

Digital images captured under poor lighting conditions often appear dark, causing important details to become hidden. Image enhancement techniques are used to improve such images by increasing their brightness and making low-intensity regions more visible. Grey level transformations, especially gamma correction with a gamma value less than one, provide an effective way to brighten dark images while maintaining natural image quality.

Explanation

The suitable transformation is **Gamma Correction** ($\gamma < 1$).

Formula:

$$s = c \times r^\gamma$$

Where:

- $\gamma < 1$

Application

- Brightens dark regions.
- Reveals hidden details.
- Improves visibility.

Justification

Gamma values less than 1 expand low-intensity values without over-brightening the image.

Conclusion

Gamma correction successfully enhances dark images by increasing the brightness of low-intensity pixels without excessively affecting brighter regions. As a result, hidden details become more visible, improving both the appearance and usefulness of the image. This technique is commonly used in surveillance systems, medical diagnosis, photography, and scientific image analysis.

3. An image has poor contrast. Explain a suitable histogram-based technique.

Introduction

Contrast plays a major role in determining the quality and clarity of an image. Images with poor contrast have a narrow range of intensity values, making objects difficult to distinguish from the background. Histogram-based enhancement techniques improve the distribution of pixel intensities, resulting in a clearer and more visually appealing image. Histogram Equalization is one of the most widely used methods for enhancing image contrast.

Explanation

The suitable method is **Histogram Equalization**.

Working

- Compute image histogram.
- Calculate cumulative distribution function (CDF).
- Redistribute pixel intensities.
- Spread intensity values across the full range.

Application

- Medical imaging
- Satellite images
- Digital photography

Justification

Histogram Equalization increases contrast by utilizing the full intensity range.

Conclusion

Histogram Equalization improves image quality by spreading intensity values over the full available range, thereby increasing the contrast between different regions of the image. This enhancement reveals hidden details and makes the image easier to interpret. Because of its simplicity and effectiveness, it is widely applied in medical imaging, remote sensing, industrial inspection, and digital photography.

4. A low-contrast image requires uniform intensity distribution.

Introduction

A low-contrast image usually contains pixel values concentrated within a small intensity range, resulting in poor visibility of important features. Uniform intensity distribution helps improve image quality by making full use of the available grey level range. Histogram Equalization is a commonly used image enhancement technique that redistributes intensity values uniformly, thereby improving both visual quality and information content.

Explanation

The best method is **Histogram Equalization**.

Working

- Calculate histogram.
- Obtain cumulative histogram.
- Map old pixel values to new intensity values.
- Produce nearly uniform intensity distribution.

Application

Used in:

- Medical images
- Remote sensing
- Security cameras

Justification

Equalization spreads pixel values uniformly, increasing contrast.

Conclusion

Histogram Equalization effectively transforms a low-contrast image into one with better visibility and improved intensity distribution. By redistributing pixel values across the entire intensity range, it enhances object boundaries and fine details. This technique is widely used in applications such as medical imaging, satellite image analysis, document processing, and machine vision.

5. An image shows uneven brightness across regions.

Introduction

Uneven brightness or non-uniform illumination is a common problem in digital images caused by varying lighting conditions or sensor limitations. Some regions may appear too bright while others remain dark, making image interpretation difficult. Adaptive Histogram Equalization, particularly Contrast Limited Adaptive Histogram Equalization (CLAHE), enhances each region independently, producing a more balanced and natural-looking image.

Explanation

The suitable enhancement technique is **Adaptive Histogram Equalization (AHE/CLAHE)**.

Working

- Divide image into small blocks.
- Apply histogram equalization separately.
- Combine the enhanced blocks.

Application

- Medical imaging
- Face recognition
- Vehicle number plate recognition

Justification

Local enhancement corrects brightness differences better than global enhancement.

Conclusion

Adaptive Histogram Equalization effectively corrects non-uniform illumination by enhancing local image regions instead of the entire image at once. This approach improves local contrast, reveals hidden details, and produces consistent brightness throughout the image. It is widely used in medical diagnosis, surveillance systems, remote sensing, and computer vision because of its superior performance under varying lighting conditions.

6. A noisy image contains random intensity variations.

Introduction

Noise is an unwanted variation in pixel intensity that degrades image quality and reduces the accuracy of image analysis. It may occur due to sensor errors, transmission problems, or environmental conditions during image acquisition. Smoothing filters are commonly used to reduce noise before performing further image processing operations. Among these, the Mean Filter is one of the simplest and most widely used techniques for reducing random noise.

Explanation

The suitable smoothing filter is the **Mean (Average) Filter**.

Working

- Replace each pixel with the average value of neighbouring pixels.

Application

- Removes random Gaussian noise.
- Smooths the image.

Justification

Averaging neighbouring pixels reduces random intensity variations.

Conclusion

The Mean Filter reduces random intensity variations by replacing each pixel with the average value of its neighbouring pixels. This smoothing process improves image quality by reducing noise, although it may slightly blur image edges. Despite this limitation, the Mean Filter remains an effective preprocessing technique in applications where reducing random noise is more important than preserving sharp edges.

7. An image requires noise reduction without significant loss of details.

Introduction

Noise should be removed while preserving In many image processing applications, it is important to remove noise while preserving important image features such as edges and fine details.

Conventional smoothing techniques often blur edges along with noise, reducing image quality.

The Median Filter is a non-linear filtering technique specifically designed to remove noise while maintaining edge information, making it highly suitable for many practical applications. Image edges.

Explanation

The suitable filter is the **Median Filter**.

Working

- Sort neighboring pixel values.
- Replace the center pixel with the median value.

Application

- Salt-and-pepper noise removal.
- Medical images.
- Digital photographs.

Justification

Median filtering removes noise without blurring edges significantly.

Conclusion

The Median Filter effectively removes impulse noise while preserving image edges and important structural details. Unlike averaging filters, it replaces noisy pixels without significantly affecting neighboring image features. This balance between noise removal and detail preservation makes the Median Filter one of the most preferred preprocessing techniques in medical imaging, object detection, and digital photography.

8. An image appears blurred after smoothing.

Introduction

Smoothing Although smoothing filters effectively remove noise, they often reduce image sharpness by weakening edges and fine details. Sharpening techniques are therefore applied to restore image clarity and improve the visibility of important features. The Laplacian Filter is a widely used sharpening operator that enhances rapid intensity changes, making image edges more distinct. Noise but may reduce sharpness.

Explanation

The suitable sharpening method is the **Laplacian Filter**.

Working

- Detect rapid intensity changes.
- Add the Laplacian result back to the original image.

Application

- Medical images.
- Satellite images.
- Industrial inspection.

Justification

The Laplacian enhances edges and restores lost details.

Conclusion

The Laplacian Filter restores image sharpness by highlighting edges and fine details that may have been lost during smoothing. It improves the visual quality of the image and increases the accuracy of subsequent image analysis. Due to its effectiveness, it is commonly used in medical imaging, industrial inspection, remote sensing, and computer vision applications.

9. Fine details in an image are not clearly visible.

Introduction

Fine details and edges contain high-frequency information that is essential for image interpretation. When these details become weak or unclear, the image appears blurred and important features may be lost. High-pass filtering is an image enhancement technique that emphasizes high-frequency components while suppressing slowly changing intensity values, thereby improving image sharpness.

Explanation

The suitable filtering technique is the **High-Pass Filter**.

Working

- Removes low-frequency components.
- Preserves high-frequency components.
- Enhances edges and fine details.

Application

- Fingerprint recognition.
- Document enhancement.
- Medical imaging.

Justification

High-pass filtering increases the visibility of fine structures.

Conclusion

High-pass filtering enhances fine image details by emphasizing edges and other high-frequency components. The resulting image appears sharper, making small structures and object boundaries easier to identify. This technique is widely used in fingerprint recognition, document processing, satellite image enhancement, and medical image analysis.

10. A noisy image must be pre-processed before further analysis.

Introduction

Image preprocessing is an essential step in digital image processing because it improves image quality before higher-level operations such as segmentation, feature extraction, and object recognition. Noisy images often reduce the accuracy of these operations by introducing unwanted pixel variations. Applying an appropriate spatial filtering technique, such as the Median Filter, helps remove noise while preserving important image structures.

Explanation

The suitable spatial filtering method is the **Median Filter**.

Working

- Select a neighbourhood.
- Arrange pixel values.
- Replace the centre pixel with the median value.

Application

- Medical imaging.
- Face recognition.
- OCR.
- Object detection.

Justification

Median filtering removes impulse noise while preserving image edges, making it suitable for preprocessing.

Conclusion

The Median Filter is an effective preprocessing technique because it removes noise without significantly affecting image edges and fine details. By improving image quality before further analysis, it increases the accuracy and reliability of subsequent image processing tasks. Its simplicity, efficiency, and ability to preserve important features make it one of the most widely used spatial filters in real-world image processing applications.